Marketing Automation Architecture Design

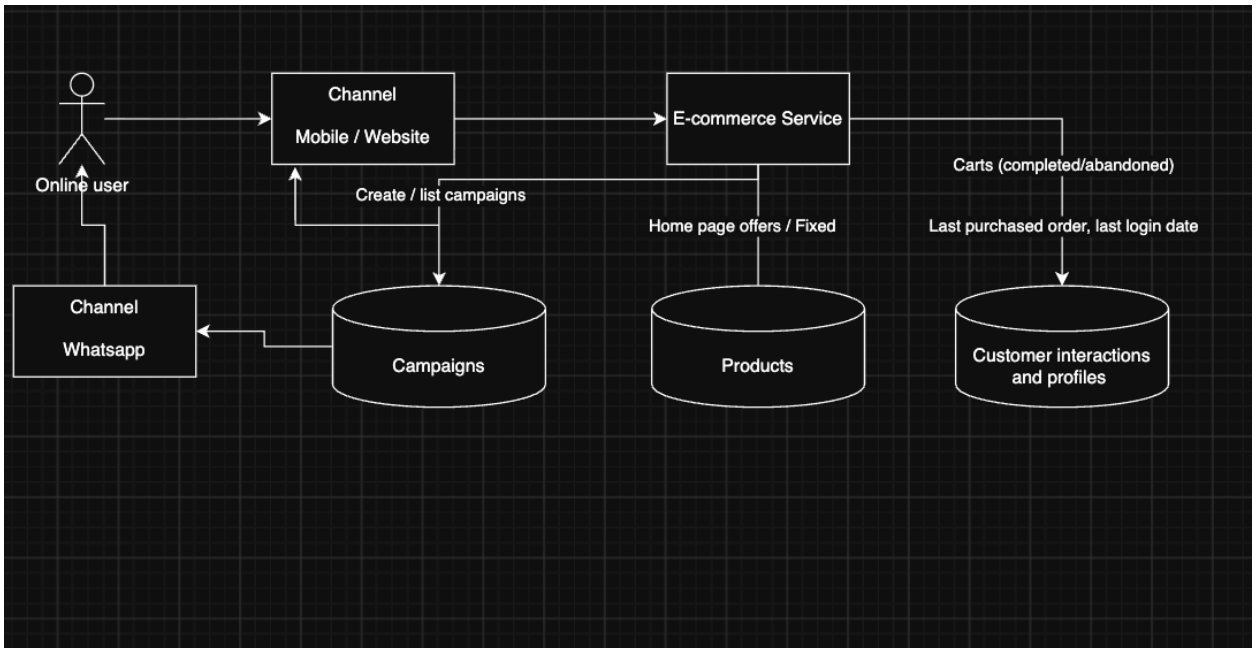
Problem statement/pain points

- Homepage / Products page Personalization. (feature stores)
    - Static offers, we need to personalize it per customer or segment.
- Next-best action
    - Abandon cart.
    - Potential churned customers.
- New users (cold start).
- Agentic Marketing - Campaign Automation.
    - Generate new marketing campaign by Agentic AI.

Goals/non-goals, Solution Assumptions

- Goals
    - Design personalization and Agentic Marketing system that should increase the CSAT to 80% after current 60%. By measuring customers rating and text feedback.
- Non-Goals
    - Lower refunds.
    -

As-is Architecture, Explore the Enterprise Architecture



Assumptions

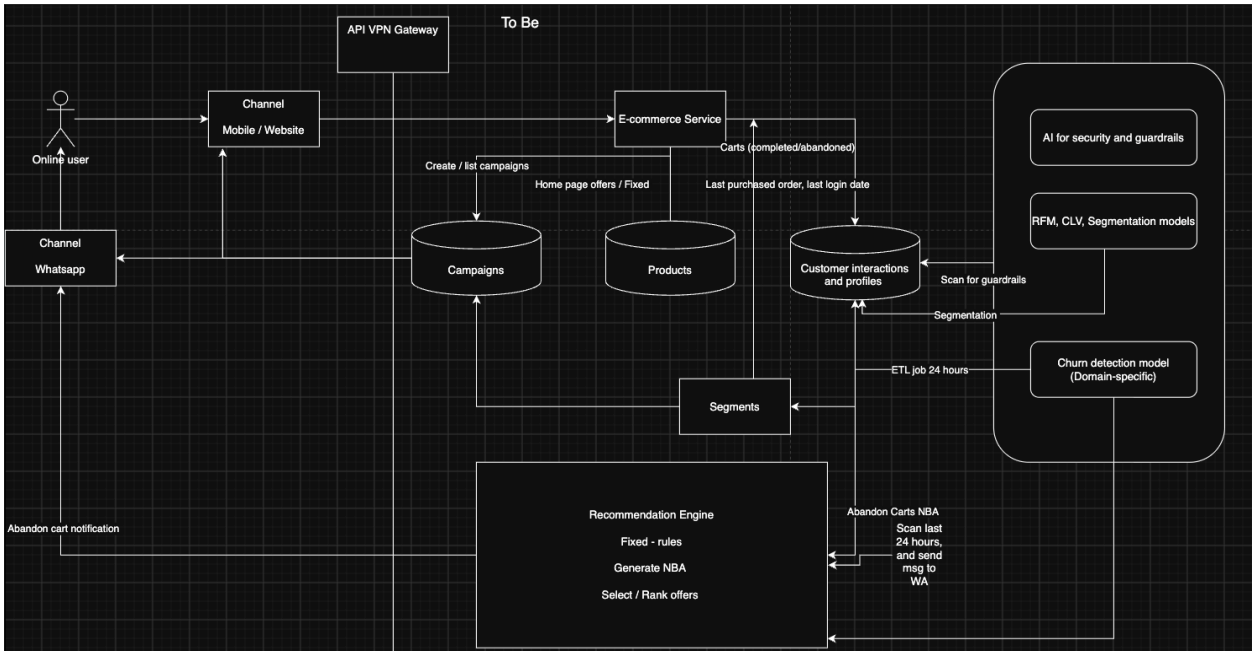
- No of concurrent users based on last 6 months (50 users).
- Seasonality (Ramadan, Holidays, …) Peak hours. 100 users.

Architecture principles, key decisions and trade-offs

- Data and compliance
    - Data should be stored in-country (on transit, on disk).
    - GDPR.
    - PDPL PII.
    - AI Guardrails to achieve clean text and limit the AI errors. Prompt injections.
- Security
    - All internal systems should be behind VPN.
    - All external systems should be authenticated and use https.

Context Viewpoint

Only Context



Components description (TBD)

- Component 1: describing it's job, and major features it's handling, also the major logical components inside, 2-3 lines per component.

Deployment Viewpoint

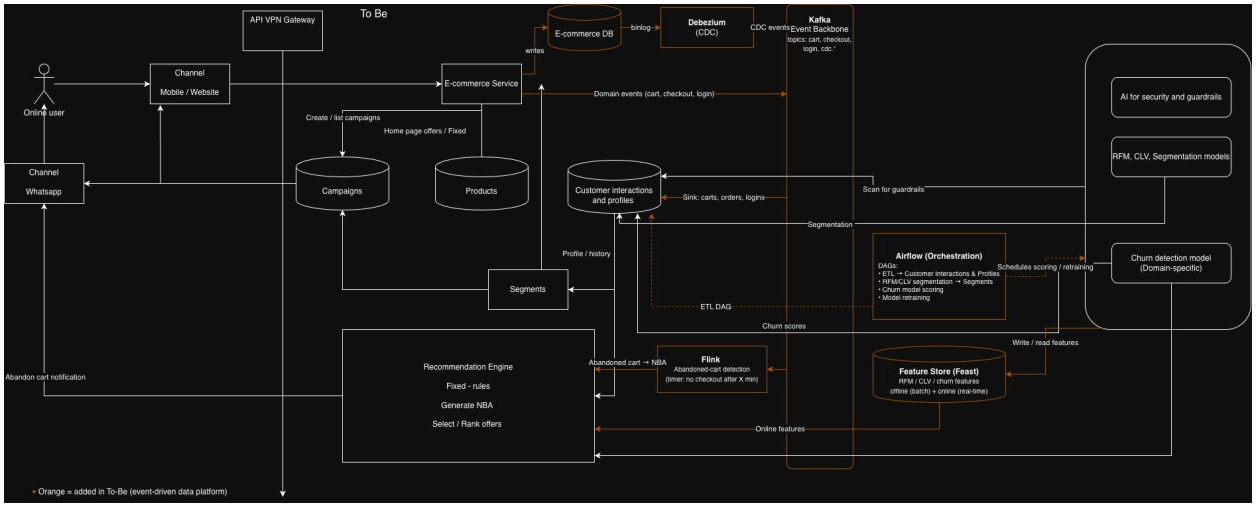
Capacity Planning

- For each service
    - 1 Node, 2 pods for every node, 1-core vCPU limit 2 vCPU / 2 GiB, limit 4 GiB.
- API / VPN Gateway added to handle (authentication/auth, rate limiting, routing/discovery, load balancing, observ. E.g. logging, metrics, ..).
- DR very recommended to be there (e.g. main DC is Riyadh, and DR at Al-khobar).

Data Viewpoint

Data Flow/Pipelines (TBD, sequence or activity diagrams)

- Customer segmentation.
- Transactions data ingestion.
- AI Security and guardrails.
- Churn-prediction.
- Abandon carts.



Connections table (TBD)

Source	Destination	Protocol	Port(s)	Authentication	Exposure (pub/private VPN)
Mobile app	E-comerce service	HTTPS (TCP) through gateway	443	JWT, TLS 1.2+	Public with Gateway authen

Data Policy

- Retention policy: all customer's data should be kept for at least 5 years.

API & Data Contracts

Endpoints

Method	Endpoint	Description
POST/GET	/v1/resource	Create new …

Data mapping / contracts
(Flink/Kafka/…. Data contracts, between two systems or more)...

Operation Viewpoint

Service Level Objectives

Perform load testing on pre-prod for the following

- Availability: 99.8%
- P50 Latency: <100ms
- P95 Latency: 500ms
- Throughput: 1000 RPS

Observability

- Grafana: Utilization metrics, logs and traces, plus dashboards, alerts and incidents.
- LangFuse: AI/ML metrics, llm as a judge, evals, drifting…
- Loki: central logs.
- Prometheus: alerts, send to channels (slack, teams, email, …).

Security Clearance

- Fill in the security survey, and get the security team sign-off.